

Loan Default Prediction using Machine Learning Models



Machine Learning Capston Project

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OVERVIEW

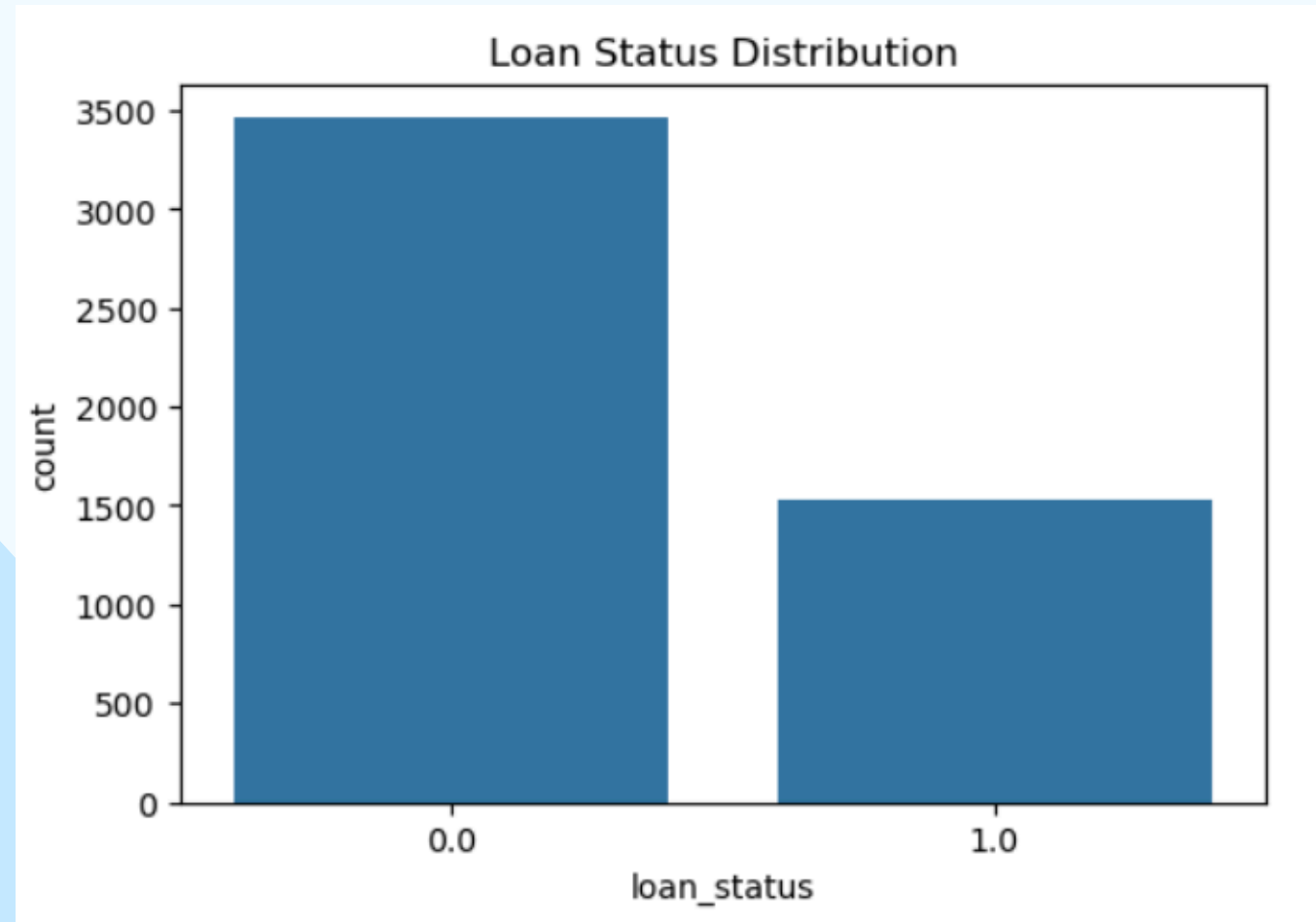
Financial institutions face losses when borrowers fail to repay loans. Early identification of potential defaulters can improve lending decisions and reduce financial risk. This project focuses on predicting loan default using machine learning techniques. Different models such as Logistic Regression, Decision Tree, Random Forest, and XGBoost were used to analyze borrower and loan-related information. Data preprocessing, feature engineering, and model evaluation were performed to identify the best prediction model. The goal of this project is to help improve loan risk assessment and decision-making.

DATASET DESCRIPTION

The dataset used in this project contains information related to borrowers and their loan details. It includes features such as age, income, home ownership, loan amount, loan interest rate, loan grade, and credit history length. The target variable is loan status, which indicates whether a borrower is likely to default or not default on a loan.

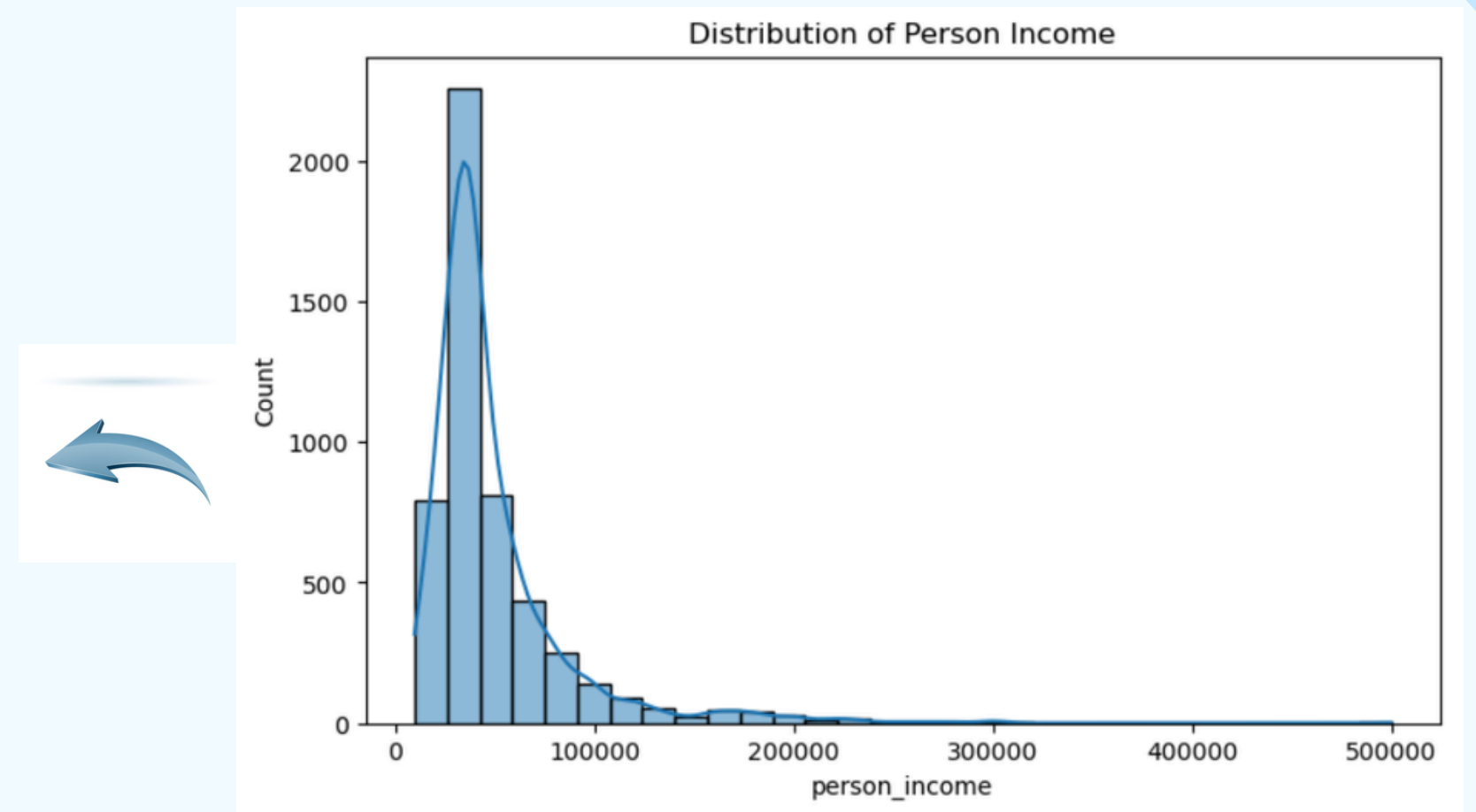
EXPLORATORY DATA ANALYSIS (EDA)

EDA was performed to understand the dataset and identify important patterns related to loan default prediction.

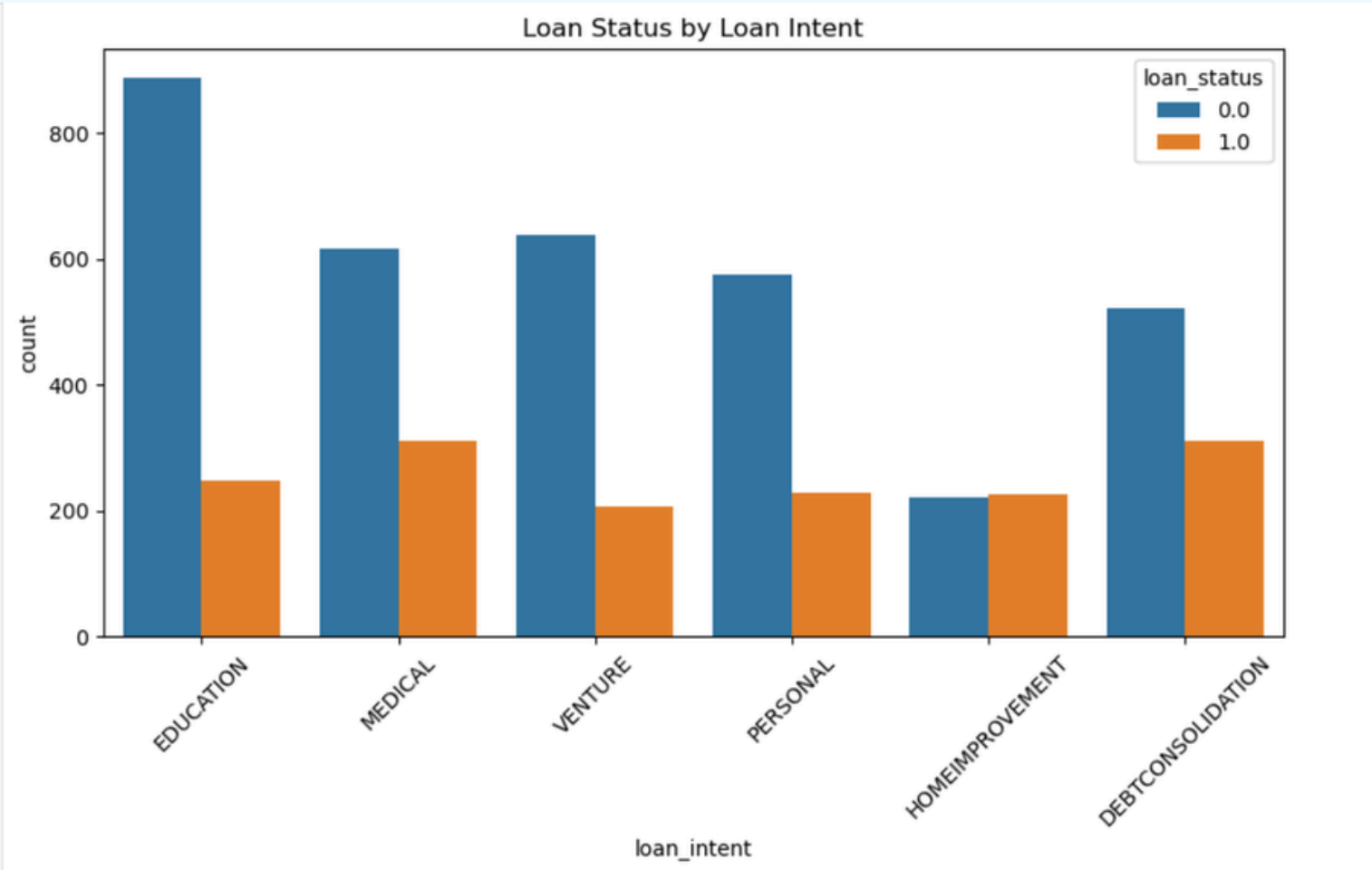


Most borrowers belong to the non-default category that is (0) while fewer borrowers belong to the default category that is (1). This indicates that the dataset contains more successful loan repayments than defaults.

Income distribution appears to be right-skewed which indicating some borrowers have significantly higher incomes than others.

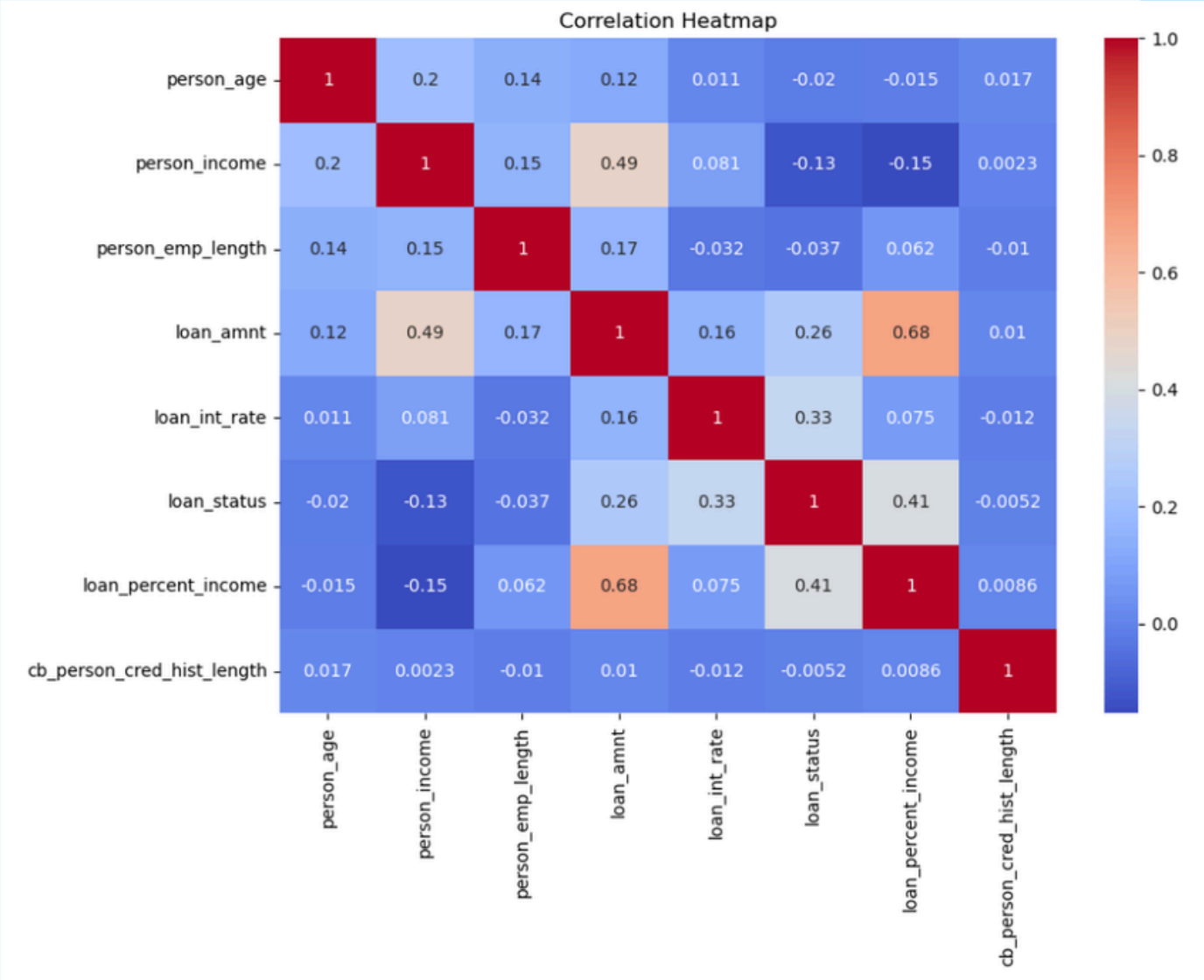


EXPLORATORY DATA ANALYSIS (EDA)



Borrowers applying for Education loans have the highest number of non-default cases while categories such as Medical and Debt Consolidation show comparatively higher default counts.

The correlation heatmap indicates that some features such as loan_percent_income and loan_int_rate are more associated with loan status, while most other variables show weak correlations.



MODEL PERFORMANCE & FEATURE COMPARISON

	Model	Accuracy	Precision	Recall	F1 Score
0	Logistic Regression	0.7948	0.6427	0.7724	0.7016
1	Decision Tree	0.8739	0.7853	0.8205	0.8025
2	Random Forest	0.9179	0.9197	0.8077	0.8601
3	XGBoost	0.9089	0.8932	0.8045	0.8465



Among all the models, Random Forest achieved the best overall performance with the highest Accuracy, Precision and F1 Score. Logistic Regression showed comparatively lower performance. Although Decision Tree showed the highest Recall, indicating strong ability to identify defaulters but overall, Random Forest and XGBoost performed better than the other models.

	Feature	Logistic Regression	Decision Tree \
3	loan_grade	0.839777	0.200692
11	person_home_ownership_RENT	0.837880	0.074452
6	loan_percent_income	1.354149	0.257981
13	loan_intent_HOMEIMPROVEMENT	0.343925	0.028117
16	loan_intent_VENTURE	0.374469	0.014353
1	person_income	0.773334	0.189765
9	person_home_ownership_OTHER	0.049268	0.002774
10	person_home_ownership_OWN	0.133871	0.006243
12	loan_intent_EDUCATION	0.410559	0.019877
0	person_age	0.130288	0.049366
15	loan_intent_PERSONAL	0.193448	0.009657
2	person_emp_length	0.046106	0.045536
14	loan_intent_MEDICAL	0.072369	0.006651
8	cb_person_cred_hist_length	0.063322	0.016512
5	loan_int_rate	0.211538	0.045886

	Random Forest	XGBoost
3	0.148160	0.193889
11	0.055033	0.173468
6	0.224213	0.119163
13	0.027213	0.069590
16	0.012221	0.066601
1	0.160590	0.053554
9	0.001346	0.050238
10	0.012183	0.049744
12	0.015878	0.044245
0	0.043478	0.034406
15	0.008683	0.028305
2	0.049637	0.026610
14	0.008059	0.025423
8	0.022998	0.024760
5	0.097081	0.016626



In this loan grade, loan percentage income, borrower income, home ownership status, and loan intent categories appeared as important predictors across multiple models. This suggests that borrower financial condition and credit quality strongly affect loan default prediction.

HYPERPARAMETER TUNING RESULTS & PERFORMANCE ANALYSIS

Accuracy: 0.913913913913914					
	precision	recall	f1-score	support	
0.0	0.91	0.97	0.94	687	
1.0	0.92	0.80	0.85	312	
accuracy			0.91	999	
macro avg	0.91	0.88	0.90	999	
weighted avg	0.91	0.91	0.91	999	
RANDOM FOREST					

Accuracy: 0.918918918918919					
	precision	recall	f1-score	support	
0.0	0.91	0.97	0.94	687	
1.0	0.93	0.80	0.86	312	
accuracy			0.92	999	
macro avg	0.92	0.89	0.90	999	
weighted avg	0.92	0.92	0.92	999	
XGBOOST					

Key Observations

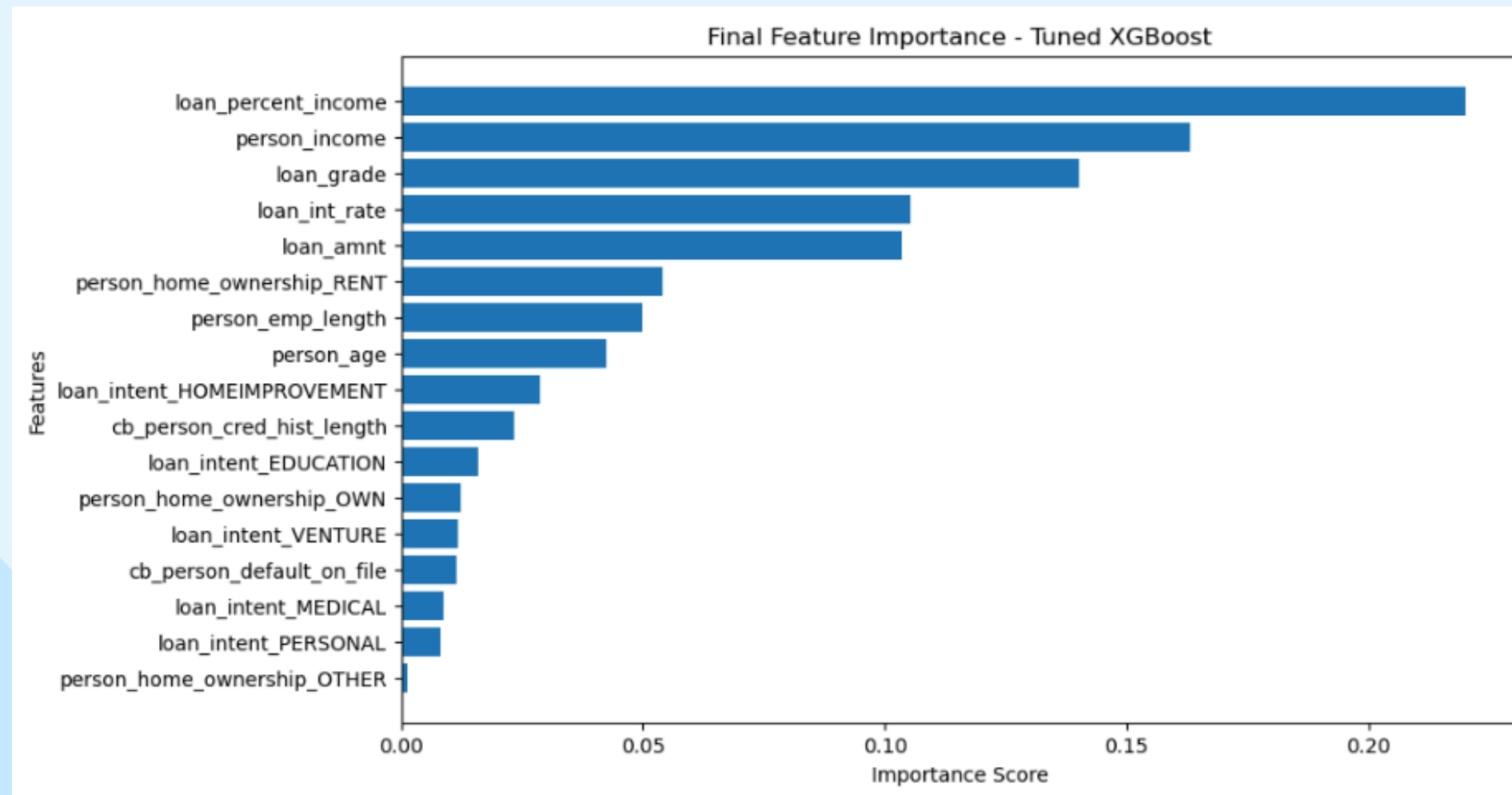
- Hyperparameter tuning improved the performance of both Random Forest and XGBoost models.
- Both models achieved strong results with Accuracy above 91%, showing reliable loan default prediction capability.
- XGBoost slightly outperformed Random Forest with higher Accuracy (91.9%), Precision (93%) and F1 Score (86%).
- Both models achieved similar Recall (80%), indicating good ability to identify potential loan defaulters.
- Based on overall performance balance, Tuned XGBoost was selected as the final recommended model for default risk prediction.

Final Model Insights & Recommendation

Key Insights

Feature importance analysis of the final tuned XGBoost model showed that loan percentage income, person income, loan grade, interest rate, loan amount, and home ownership status were the strongest factors influencing loan default prediction.

Borrowers with higher loan burden, lower income stability, weaker loan grades, and higher interest rates showed greater default risk.



Recommendation

- Use the Tuned XGBoost model to support loan approval decisions.
- Give extra attention to applicants with high loan amount compared to income, lower loan grades, and higher interest rates.
- The model can help banks detect risky borrowers early, reduce default losses, and make better lending decisions.

THANKYOU